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The palaeomagnetism of Jersey volcanics and dykes, and the Lower Palaeozoic apparent polar wander path for Europe

B. A. Duff* *Department of Earth Sciences, University of Leeds, Leeds LS2 9JT*

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Summary. Stable natural remanent magnetization (NRM) in the Jersey Volcanics and in a single rhyolite dyke was probably acquired during the Cambrian before folding of the volcanics in the Cadomian Orogeny. After dip correction, the volcanics yield a palaeomagnetic pole at 323° E, 52° N (dp = 33°, dm = 35°). In Jersey dolerite dykes three groups of stable NRM directions are recognized, with palaeomagnetic poles at 248° E, 26° N (dp = 10°, dm = 20°), 339° E, 1° S (dp = 9°, dm = 12°), and 336° E, 31° S (dp = 5°, dm = 9°). Comparison with the European apparent polar wander path implies that stable NRM in these groups was acquired respectively during Late Precambrian or early Cambrian, Siluro-Devonian and middle Carboniferous time. The stable NRM of the Jersey lamprophyre dykes yields a palaeomagnetic pole at 322° E, 16° N (dp = 31°, dm = 38°) and is probably of Silurian or Devonian age.

These palaeomagnetic poles and other new data determined by the author for the Armorican Massif can be fitted to a common apparent polar wander path for Europe, and this implies that the basement of Lower Palaeozoic Europe extended from the Baltic Shield at least as far south as the Armorican Massif. The Hercynian Orogeny in these parts of Europe was therefore probably intracratonic. This polar wander path implies that in early Cambrian time the pole did not move significantly relative to Europe, but that this was followed by a large middle to late Cambrian polar shift which corresponded to rapid drift of Europe across the South Pole.

1 The Jersey Volcanics

1.1 INTRODUCTION

Acid and intermediate volcanics (Mourant 1933) crop out mainly in north-east Jersey, and were sampled as part of a palaeomagnetic survey of the island to improve the Lower Palaeozoic palaeomagnetic record of western Europe. The basal volcanic unit, the Jersey Andesite

* Present address: Department of Geology, The Australian National University, Box 4, PO, Canberra, ACT, 2600, Australia.

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Formation, overlies Upper Brioverian metasediments (Mourant 1933) and is overlain, probably unconformably, by rhyolites, tuffs and ignimbrites.

The age of the Jersey Andesite Formation has been estimated at 533 ± 16 Myr (middle Cambrian) by Rb–Sr whole rock dating ($\lambda Rb = 1.39 \times 10^{-11}$ yr; Duff 1978a). The acid volcanics, too, are probably not much younger than middle Cambrian because they are overlain unconformably by the Rozel Conglomerate which is considered to be Cambrian or Ordovician (e.g. Renouf 1974).

Palaeomagnetic collections of the acid volcanics were made at several rhyolite, tuff and ignimbrite sites along the well-exposed section on the east coast north of Gorey, and at four sites on or near the north coast (Fig. 1). A rhyolite dyke (site J9) was sampled east of Mont Orgueil Castle. The Jersey Andesite Formation was sampled on the eastern shore of St Aubin's Bay (Fig. 1).

1.2 NRM OF THE JERSEY VOLCANICS

Intensities of total NRM for the 13 rhyolite, tuff and ignimbrite sites ranged from about 10^{-4} to 6×10^{-2} Am^{-1}. The rhyolite dyke (site J9) and the andesites possessed much larger intensities of up to 1.5×10^{-1} and 5×10^{-1} Am^{-1} respectively. Site mean directions of total NRM before dip correction are either very steep or variably inclined south-westerly downward directions (Fig. 2(a)); correction for structural dip results in a slightly larger scatter of directions (Fig. 2(b)).

With two exceptions (sites SC1 and SC2) all acid volcanic sites and the andesite site are stable against alternating field (a.f.) demagnetization. Typically, a quarter of the original NRM remains after demagnetization in peak fields from about 100 to 160 mT, and directions of NRM are consistent over this range of applied fields (e.g. dip-corrected directions

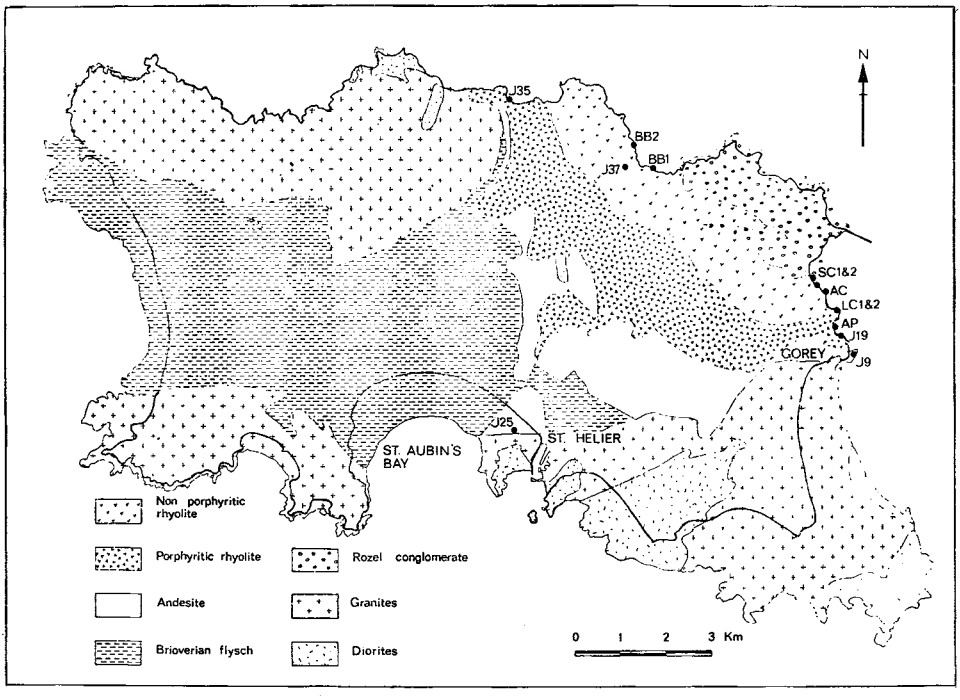


Figure 1. The geology of Jersey (Long. 358° E, Lat. 49° N).

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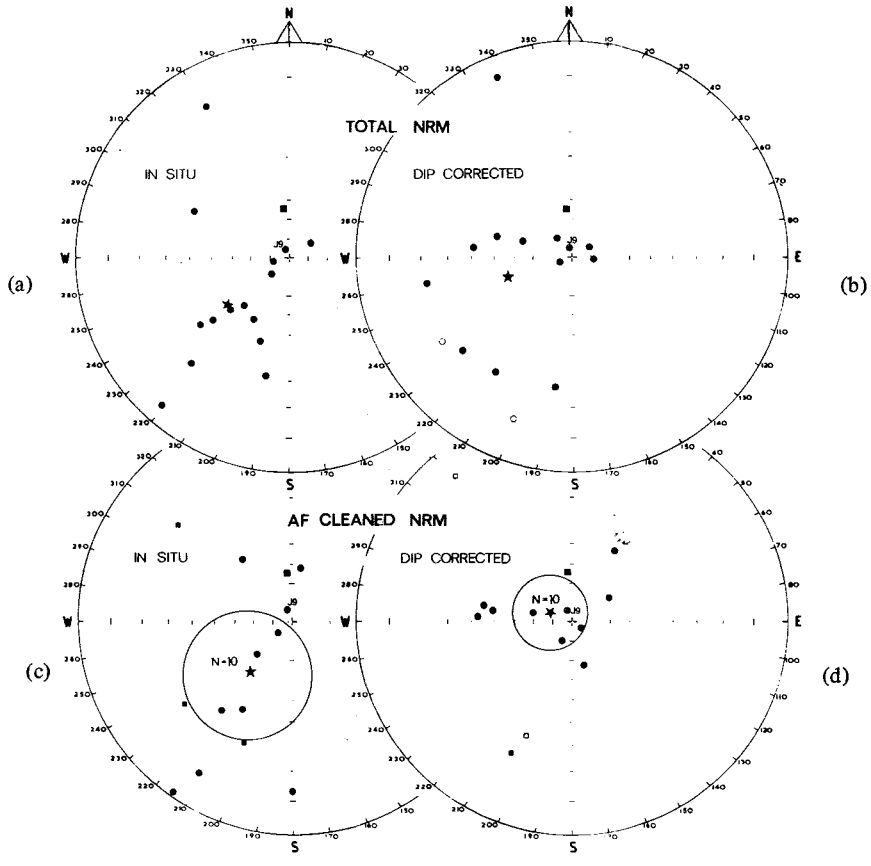


Figure 2. *In situ* and dip-corrected site mean directions of NRM for the Jersey Volcanics before (a) and (b) and after (c) and (d) optimum a.f. cleaning. The corresponding group mean directions (stars) and their circles of 95 per cent confidence (c) and (d) are also shown. The rhyolite dyke (J9) direction, directions of sites with scattered, multi-component NRM (small squares), and the present field direction (large square) are indicated. Closed symbols: downward directions; open symbols: upward directions. Stereographic projections.

Fig. 3(a)–(c)). This stability is consistent with thermomagnetic and optical results which indicate that haematite is the main remanence carrier.

The rhyolite dyke and andesites are less resistant to a.f. demagnetization (Fig. 3(d)) than the acid volcanics, and this is consistent with indications that magnetite carries most of the NRM in the dyke and andesites. Three-quarters of the original dyke NRM is demagnetized in peak fields of 30 mT. Most acid volcanic sites and the rhyolite dyke yield steep endpoints (dip corrected, Fig. 3(b)–(d)).

Some specimens from sites J19, J35, AC and LC gave a.f. demagnetization trends suggestive of multi-component NRM (e.g. Fig. 3(e)). The *in situ* NRM directions of separate specimens in these sites after demagnetization in progressively higher fields converge toward a south-westerly direction with an intermediate, downward inclination which corresponds to a hard component acquired after folding of the volcanics. This direction is close to the Devonian direction of south-west Europe.

Thermal demagnetization of representative specimens from the acid volcanic sites with stable single component NRM (e.g. Fig. 3(f) and (g)) indicates that the NRM is distributed

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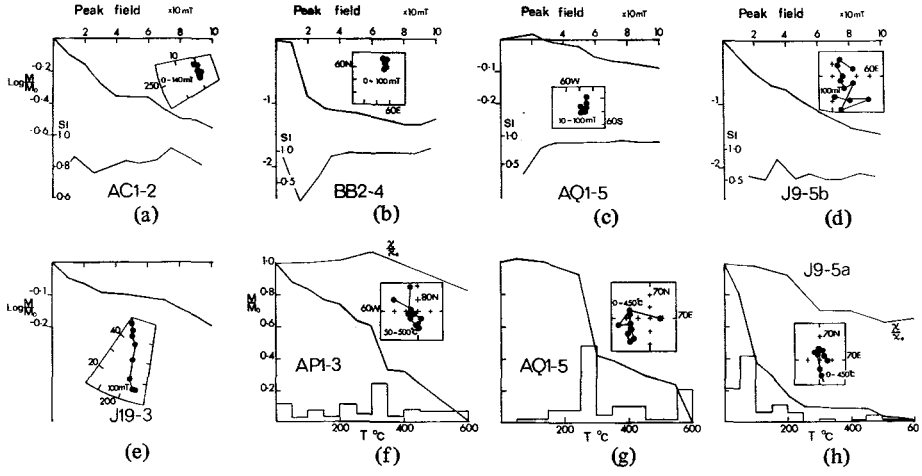


Figure 3. (a)–(e) Progressive a.f. demagnetization of the NRM in sites from the Jersey Volcanics. Corresponding directional changes (portions of stereographic projection) and the variation of stability indices with peak field are also shown. (f)–(h) Thermal demagnetization of rhyolite (AP1), tuff (AQ1) and rhyolite dyke (J9), with corresponding directional changes, blocking temperature spectra and normalized variation of susceptibility indicated. All directions downward and dip-corrected.

in the blocking temperature range 0 to about 600°C, and that the directions of NRM are consistent up to about 450°C or 500°C and close to those isolated by a.f. demagnetization. The magnetite-bearing rhyolite dyke from Mont Orgueil has a distributed blocking temperature spectrum with the bulk of single component NRM in the range 0–250°C (Fig. 3(h)), and consistent NRM directions up to 450°C similar to the dip-corrected directions isolated in most acid volcanic sites by both a.f. and thermal demagnetization.

1.3 ANALYSIS OF NRM DIRECTIONS

Thermal and a.f. demagnetization show that a minority of sites possess two components of NRM, one acquired before and one after folding. At the majority of sites, where NRM is single-component it is therefore important to determine if it was acquired before or after folding.

In this respect, the direction of stable NRM in the rhyolite dyke is important for the interpretation of the NRM directions of the volcanics. This dyke is of very similar composition to the acid volcanic flows which were sampled, and may have been a contemporaneous feeder for those flows (Mourant 1933). Because the dyke has not been folded or tilted since emplacement, its NRM direction is unambiguous. In contrast with most *in situ* site mean results, the dip-corrected mean directions of several sites (Fig. 2) are close to the steep direction characteristic of the dyke. This implies that at least for many sites, NRM was acquired before folding as a single component with steep inclination.

Sites with low stability against a.f. demagnetization and sites with many specimens containing variable multi-component NRM were excluded from the overall analysis. The dip-corrected NRM directions from the remaining 10 sites (Fig. 2(d)) are distributed about a mean $D = 292°$, $I = 78°$ with precision $\kappa = 7.7$. A two-tier analysis (Watson & Irving 1957) of the dip-corrected NRM directions of the 10 sites implies a significant between-site scatter at the 99 per cent probability level; between-site scatter ($\beta = 9.4$) is the main source of overall disperson.

It is likely that a large part of the between-site scatter of NRM directions in acid volcanics results from the extremely viscous nature of acid flows, as a result of penecontemporaneous

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Table 1. Mean directions of NRM and palaeomagnetic pole positions for the Jersey Volcanics and the dolerite and lamprophyre dykes of Jersey.

	N	D°	I°	R	k	α_{95}	Long.	Lat.	dp°	dm°
Jersey Volcanics	10 sites	292	78	8.8	7.6	19	323 E	52 N	33	35
	66 samples	294	79	56.7	7.0	7	324 E	53 N	12	13
Group A dolerites	5 dykes	199	16	4.9	71.2	9	336 E	31 S	5	9
Group B dolerites	6 dykes	204	57	5.9	59.2	8	339 E	1 S	9	12
Group C dolerites	7 dykes	302	16	6.4	10.5	20	248 E	26 N	10	20
Lamprophyre dykes	4 dykes	234	64	3.8	15.8	24	322 E	16 N	31	38

Abbreviations: D, I declination, inclination of NRM; R resultant of N unit vectors; k estimate of precision parameter; α_{95} semi-angle of the cone of 95% confidence about the mean direction; dp, dm semi-angle subtended by major and minor axes of ellipse of 95% confidence about the mean pole position.

folding of partially congealed lava below the blocking temperatures of its NRM. Such synvolcanic folding is common in the acid volcanics of Jersey.

Comparison of the *in situ* and dip-corrected between-site precisions for the 10 site mean directions of NRM by Cox's (1969) method indicates that the estimates of the 95 per cent probability level confidence intervals β_u, β_l about the true precision for the *in situ* directions ($\beta_u = 6.4 > \beta = 3.8 > \beta_l = 1.9$) and the dip-corrected directions ($\beta_u = 15.7 > \beta = 9.4 > \beta_l = 4.7$) overlap slightly. The fold test is therefore inconclusive at the 95 per cent probability level, although the difference between *in situ* and dip-corrected precisions favours the view that the bulk of NRM was acquired before folding, as suggested by the similarity of dip-corrected site mean directions in the volcanics with the direction of stable NRM in the rhyolite dyke.

The best estimate of NRM direction in the Jersey Volcanics is therefore the dip-corrected mean of site mean directions $\bar{D} = 292°$, $I = 78°$ ($\alpha_{95} = 19°$), which yields a palaeomagnetic pole at 323° E, 52° N ($dp = 33°$, $dm = 35°$; Table 1). The large oval of confidence about the mean pole is a consequence of dipole field properties at high latitude (see, for example, Irving 1964, Figure 4.7).

1.4 THE ORIGIN AND AGE OF NRM IN THE JERSEY VOLCANICS

The very high stabilities of NRM against a.f. demagnetization and thermomagnetic results suggest that the dominant magnetic phase and carrier of NRM in the rhyolites, tuffs and ignimbrites is haematite. In these rocks small regions of relict magnetite occur within octahedral crystals of haematite, which is arranged in a mesh texture parallel to the vestigial octahedral cleavage. These features suggest that original magnetite has been pseudomorphed by haematite during martitization. In some ignimbrite sites haematite is mainly contained within phenocrysts of amphibole or biotite.

Oxidation of biotite to haematite is common in volcanic rocks (Eugster & Wones 1962); it proceeds first – perhaps before extrusion – to magnetite and alkali feldspar, and after extrusion magnetite is oxidized, partially or completely, to haematite. Much of the haematite in the Jersey Volcanics was probably derived from magnetite during the initial cooling

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of the lava pile. The magnetite may in turn have been derived from primary ferromagnesian phenocrysts, or have crystallized independently.

The main Fe–Ti oxide in the rhyolite dyke is magnetite, which is present as discrete octahedral crystals. The presence of magnetite in the dyke in contrast to the haematite of the extrusives presumably reflects the higher oxygen fugacities of the extrusives on surface cooling.

According to the thermal demagnetization results, temperatures of about 200°C (Pullaiah *et al* 1975) over sustained periods (> 1 Myr) would have been necessary for the NRM associated with the initial formation of haematite to have been subsequently reset. However, the pre-tectonic folding age of the NRM of the volcanics suggests that the burial necessary for such remagnetization has not occurred. The age of crystallization estimated for the Jersey Andesite Formation provides a maximum age for the overlying acid volcanics. Folding of the Jersey Volcanics occurred during the Cadomian Orogeny before deposition of the Rozel Conglomerate in late Cambrian or early Ordovician times. The bulk of stable NRM was therefore probably acquired during the interval 533 Myr (middle Cambrian) to early Ordovician.

2 Jersey dolerite and lamprophyre dykes

2.1 INTRODUCTION

Palaeomagnetic collections were made of 29 dolerite dykes intrusive into the south-east granite and diorite, the south-west granite and the Brioverian metasediments, and eight lamprophyre dykes intrusive into the three granites, the south-east diorite and the Rozel Conglomerate of Jersey (Fig. 1). The ages of the dolerite dykes have not been determined, although both dolerites and lamprophyres must by younger than the granites and diorites (Late Precambrian to Ordovician). A lamprophyre which intrudes the Rozel Conglomerate has yielded a K–Ar hornblende age of 427 ± 13 Myr (Adams 1967).

2.2 NRM OF THE JERSEY DOLERITES AND LAMPROPHYRES

Dolerite and lamprophyre dyke mean directions of total NRM (Fig. 4) are scattered, although most are steep and downward pointing. Intensities of NRM in the dolerites and lamprophyres range from 10^{-3} to 10 Am^{-1}

On a.f. demagnetization in peak fields up to 100 mT, 21 of the 29 dolerite dykes yielded stable endpoints (Fig. 5). Magnetic cleaning of 18 of the 21 stable dykes yielded three distinct palaeomagnetic groupings of dyke mean directions (Fig. 4(b)). These are distinguished entirely on palaeomagnetic grounds; there are no cross-cutting relationships between the dykes in the three groups which could support the distinction as reflecting differences in relative age of the dyke groups, although group C dykes all belong to a swarm of wide, north-east-trending dykes. Dykes of group A have shallow, downward south-south-west stable NRM directions; group B dykes are magnetized steeply downwards to the south-west (although one dyke from St. Brelade has an NRM direction reversed with respect to this group); and group C dykes have shallow downward or upward north-west directions (Fig. 4). Three dykes (J10, J33 and J42) gave directions of stable NRM oblique to these groups (Fig. 5(e) and (f)).

Group A dykes intrude an older phase of the south-west granite and the south-east granite near Dicq. They are mainly east or east-north-east-trending dykes of variable thickness (40 cm to 5 m). The stable NRM of group A dykes generally resides in coercivities of remanence from 0 to 60 mT (Fig. 5(a)).

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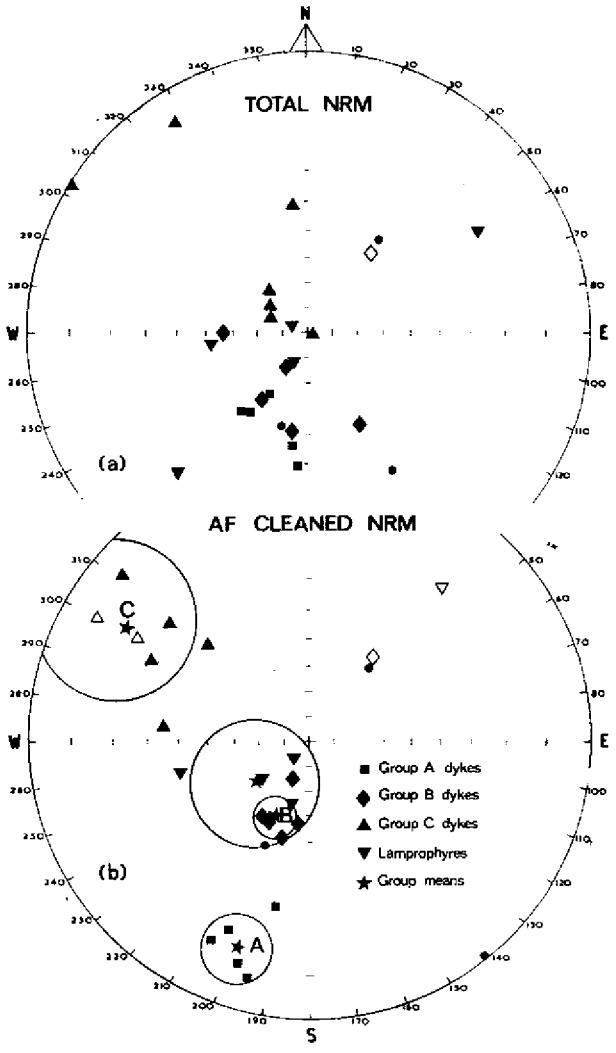


Figure 4. Jersey dolerite and lamprophyre dyke mean directions of NRM before (a) and after (b) optimum a.f. cleaning. The dolerite a.f. cleaned directions define three distinct groups (A, B and C). The corresponding mean directions of these groups and that of the lamprophyres are shown (stars) with their circles of 95 per cent confidence. Closed symbols: downward directions; open symbols: upward directions. Stereographic projections.

Dykes with group B directions have variable trends and thicknesses (40 cm to 6 m), and intrude the south-west granite and south-east diorite of Jersey. Resistance to a.f. demagnetization and directional consistency of NRM is high in peak fields up to about 100 mT (Fig. 5(b)).

Group C dykes belong to a swarm of wide, north-east-trending dykes which intrude the south-east granite. Stability of group C NRM is variable between and within dykes, and this variability may account for the large scatter in a.f. cleaned dyke mean directions (Fig. 4(b)). Resistance of the NRM of the fine-grained dyke margins to a.f. demagnetization in peak fields of 140 mT implies that the NRM is carried by single-domain magnetite or by haematite. The magnetizations removed during rotation of the resultant NRM toward group C

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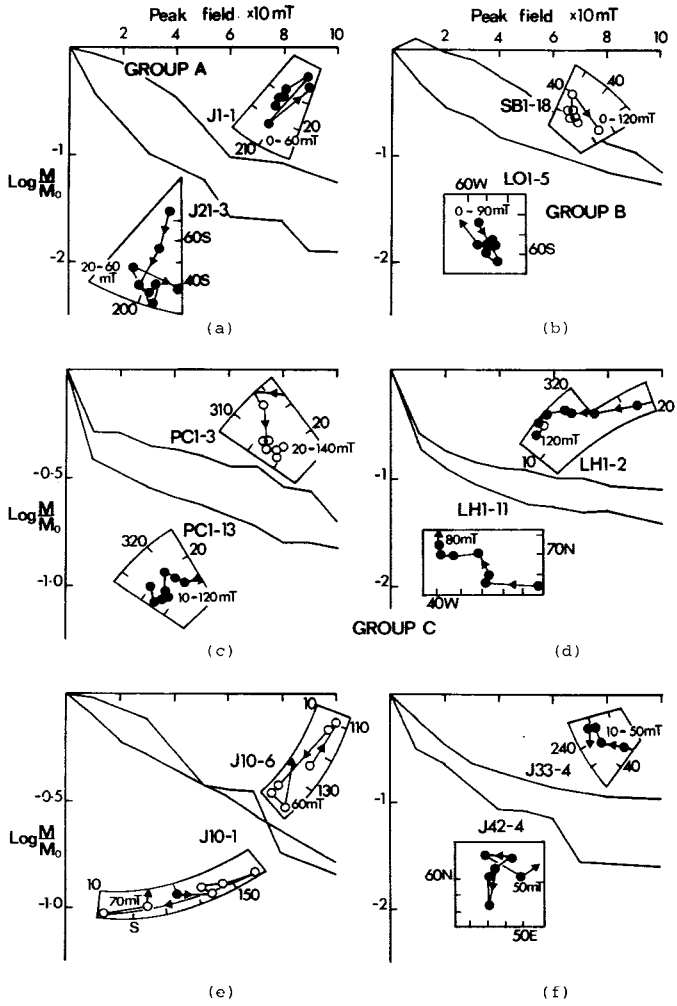


Figure 5. Progressive a.f. demagnetization of the NRM in dolerites from groups A, B and C and dykes J10, J33 and J42. Corresponding directional changes are indicated. Closed symbols: downward directions; open symbols: upward directions.

endpoints have either shallow north-north-west downward directions (Fig. 5(c) and (d)), or steep northerly directions similar to those of the Jersey granites (Duff 1978b).

Eight dolerite dykes were unstable against a.f. demagnetization and were almost completely demagnetized in low peak fields.

Five of the eight lamprophyre dykes yielded stable endpoints during a.f. demagnetization. The NRM of the remaining three was mostly demagnetized in peak fields of 10–20 mT. A lamprophyre from South Hill (SH) showed a large decay of NRM in peak fields less than 10 mT – perhaps owing to demagnetization of IRM – but considerable resistance to further demagnetization (Fig. 6(a)). The NRM of a lamprophyre from Point des Pas (J4) also shows directional consistency and stability against peak fields greater than about 10 mT (Fig. 6(b)). The NRM of the lamprophyre dykes which intrude the south-west granite (J22 and J27) is demagnetized in peak fields greater than about 40 mT, but at lower demagnetizing fields NRM directions are consistent and similar to the endpoints isolated in the more stable dykes

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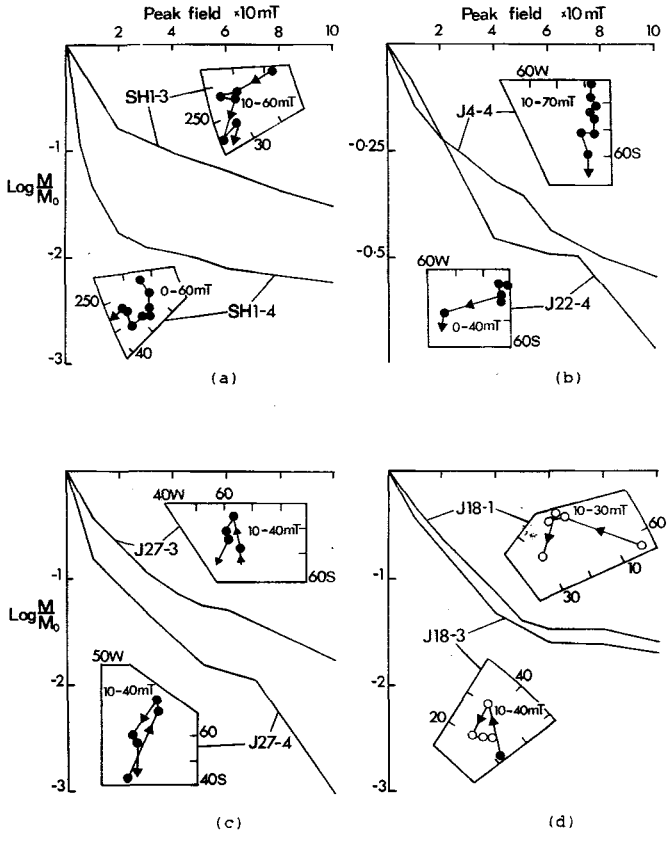


Figure 6. Progressive a.f. demagnetization of the NRM in the Jersey lamprophyre dykes. Corresponding directional changes are indicated. Closed symbols: downward directions; open symbols: upward directions.

(Fig. 6(b) and (c)). Specimens from a mica lamprophyre (J18) which intrudes the north-west granite show an exponential decay of NRM in peak fields of 10–40 mT (Fig. 6(d)). However, the endpoint is at variance with the stable NRM directions isolated in the other lamprophyre dykes.

2.3 THERMOMAGNETIC AND OPTICAL ANALYSIS

2.3.1 Jersey dolerites

Two groups of Curie points associated with two distinct types of thermomagnetic curve ($M_s(T)$ curve) characterize the Jersey dolerites. Dykes with Curie points of 500–600°C characteristic of titanium poor or pure magnetite (Fig. 7(b) and (c)) have Néel's (1948) Q type or nearly straight heating curves. In contrast, high Curie points up to 770°C are associated with concave heating curves, and possibly reflect the presence of maghaemite (Fig. 7(a) and (d)). These characterize most group A dykes and also dykes J10, J33 and J15.

Different dykes within the three palaeomagnetic groups, and the dykes with unstable NRM have $M_s(T)$ curves that are either Q type, straight or concave; there is no consistency of $M_s(T)$ curve or Curie points within groups. This implies that the NRM groupings were not controlled by systematic differences in magnetic mineralogy.

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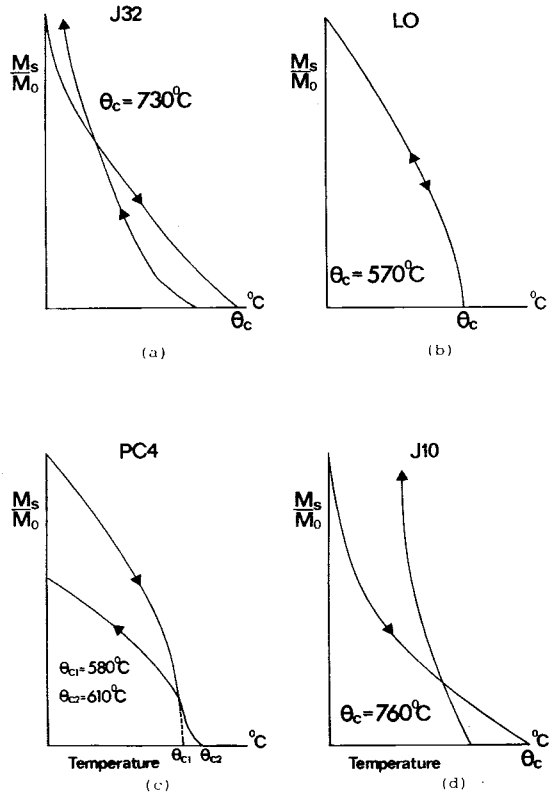


Figure 7. Thermomagnetic curves for Jersey dolerites.

Reflected light observations indicate that light brown or grey magnetite is the most common magnetic phase in the Jersey dolerites. It is present as discrete homogeneous crystals with few or no observable ilmenite or haematite lamellae, and therefore belongs to the lowest oxidation classes 1 and 2 (Wilson & Haggerty 1966). The steel grey colour of the magnetites in some dykes, especially those in group C, and the relatively high Curie points suggest that the magnetites contain little titanium. The paradox of high Curie points of magnetites in basic rocks (which are expected to be high in titanium and hence have relatively low Curie points) has been discussed by Ade-Hall (1964), Ade-Hall, Wilson & Smith (1965) and Creer, Hedley & O'Reilly (1975) with reference to basic extrusives. Following these authors, the high Curie points of the Jersey dolerites are ascribed to submicroscopic unmixing of titanium poor magnetite. Of the dykes with Curie points characteristic of maghaemite, only J31 and J33 show alteration of magnetite to maghaemite. Maghaemite may be present as submicroscopic grains in the other dykes.

Haematite is present in minor amounts associated with the magnetites as marginal grains and patchy inclusions. In a few of the palaeomagnetically unstable dykes, ilmenite is the main oxide phase and is typically present as large homogeneous phenocrysts, some of which have hollow cores.

2.3.2 *Jersey lamprophyres*

Most lamprophyre dykes have Q type $M_s(T)$ curves with Curie points characteristic of titanium poor or pure magnetite. The mean Curie point estimated from the Q type heating

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curves is 542 ± 30°C (1σ), which corresponds to a magnetite composition of about $Fe_{2.95}$, $Ti_{0.05}$ O_4. Dykes J13 and J22 gave concave $M_s(T)$ curves with high Curie points appropriate to titanohaematite (640°C) and possibly maghaemite (730°C) respectively.

Reflected light microscopy also suggests that the dominant magnetic phase is titanium-poor magnetite. Fine haematite lamellae comprise a few per cent of the magnetites of some dykes but typically haematite is absent, and the magnetites are unoxidized and homogeneous (oxidation class 1 of Wilson & Haggerty 1966), unless submicroscopic unmixing has occurred. No magnetic phases were observed in two of the palaeomagnetically stable lamprophyres despite indications of the presence of magnetite from thermomagnetic analysis. This suggests that the stable NRM may be carried by submicroscopic magnetite.

2.4 MEAN DIRECTIONS OF NRM AND PALAEOMAGNETIC POLE POSITIONS

Mean directions of stable NRM for the three dolerite groups and the lamprophyre dykes (except J18) are shown in Fig. 4(b), and summarized in Table 1. A mean direction of NRM is given for group C dykes, although the large scatter of directions between dykes in this swarm ($\kappa = 10$; $\alpha_{95} = 20°$) is noted. The stable NRM in group C dykes is believed to be Late Precambrian to early Cambrian in age (Section 3.2), and the scatter in this group may reflect real variations in the Late Precambrian geomagnetic field, as rocks with NRM of similar age in Denmark, the Nexϕ Sandstone, also record a large between-site scatter (Prasad & Sharma 1978).

The three stable dolerite dykes with large within-dyke scatter of NRM directions, oblique to directions in the three groups, were excluded from the group analysis. The stable lamprophyre dyke, J18, has a mean direction $D = 41°$, $I = -19°$, which is significantly different at the 95 per cent probability level from the other stable dyke groups; it was also excluded from the overall analysis.

3 Palaeozoic apparent polar wander and its geological implications

3.1 THE APPARENT POLAR WANDER PATH FOR EUROPE

3.1.1 Introduction

The European Lower Palaeozoic apparent polar wander (apw) path is poorly known and published paths have been based on results mainly from Britain and Norway. The single British pole position from rocks of Cambrian age – the Lower Cambrian Caerfai Series (Briden, Irons & Johnson 1970) – differs from that from the Lower Cambrian Fen Carbonatite complex (Poorter 1972) of Norway. A controversial discrepancy between Norwegian and British Siluro–Devonian results has been explained by complex remagnetization of the British rocks (e.g. Storetvedt & Halvorsen 1968), by Lower Palaeozoic relative movements between the Baltic Shield and Britain (Briden, Morris & Piper 1973), and by reinterpretation of the magnetization ages for the Norwegian rocks (McElhinny & Briden 1971).

It is doubtful whether the Caerfai Series pole reflects a Cambrian magnetization. The location of the series close to the Hercynian front in Wales, and the position of its pole close to well-established Carboniferous poles for Europe suggest that the Caerfai Series was remagnetized during the Hercynian Orogeny.

The oldest palaeomagnetic pole positions on the British path of Briden *et al.* (1973) are therefore of Ordovician age. New data from Jersey (including the results from the Jersey volcanics and dykes described in this paper), the Armorican massif (Duff 1979a), Leicestershire (Duff 1979b), and a minor study in North Wales (Duff 1978b) are summarized in

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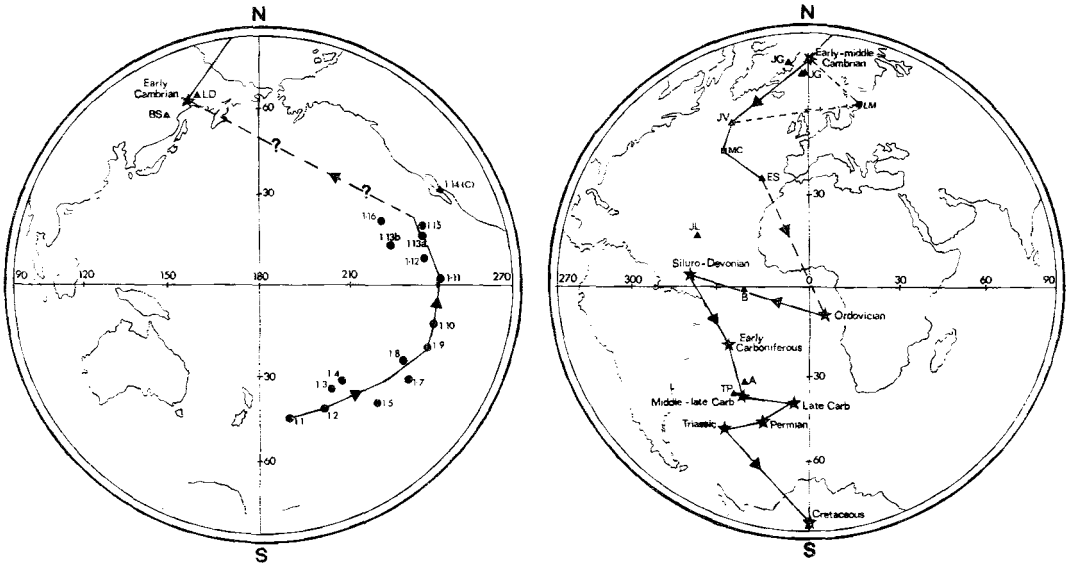


Figure 8. Suggested Late Precambrian to present apw path for Europe. The path segments joining the early Cambrian group pole with the path through the Late Precambrian poles (dots), and the Erquy Spilite Series pole (ES) with the mean Ordovician pole are tentative. The mean period poles (Table 3) are indicated by stars. Symbols for the new pole positions for the Armorican Massif and Britain (triangles) refer to Table 2. A possible alternative early to middle Cambrian path through the pole for the Longmynd sandstone (LM) is also shown. MC is the pole for the *in situ* Mona Complex (Lomax 1975). Equatorial equal area projections.

Table 2 and illustrated in Fig. 8, where they are superimposed on the south apw path for Europe, compiled from McElhinny (1973), McElhinny & Cowley (1977), J. C. Briden (private communication) and the present study (Table 2). These poles fall in a series of groups of early Cambrian, early to middle Cambrian, Ordovician, Siluro–Devonian, early Carboniferous and late Carboniferous age, for which means are illustrated in Fig. 8 and listed in Table 3. For the period Permian to present the mean poles of McElhinny (1973) were

Table 2. Summary of all palaeomagnetic results from Jersey, Brittany, Normandy (France) and Leicestershire and Wales (Britain).

	N (Sites)	D	I	R	k	α_{95}	Palaeomagnetic pole long.	lat.	(dp,dm)	Pole ref.	Age of NRM (Myr)
SE Granite, Jersey (520±4)	13	350	75	12.8	54	6	337E	77N	(9,10)	JG	Early middle Cambrian*
NW Granite, Jersey (490±15)	9	356	77	8.9	71	6	353E	73N	(7,7)	JG	" " "
SW Granite, Jersey (565±15)	11	359	77	10.9	87	5	356E	74N	(8,9)	JG	" " "
Acid and intermediate volcanics, Jersey	10	292	78	8.8	8	19	323E	52N	(33,35)	JV	533 ± 16
Dolerite Group A, Jersey	5(dykes)	199	16	4.9	71	9	336E	31S	(5,9)	A	Middle late Carboniferous*
Dolerite Group B, Jersey	6 "	204	57	5.9	59	8	339E	1S	(9,12)	B	Silurian-Devonian*
Dolerite Group C, Jersey	7 "	302	16	6.4	10	20	248E	26N	(10,20)	C	Late-Precambrian or early Cambrian?*
Lamprophyre dykes, Jersey	4 "	234	64	3.8	16	24	322E	16N	(31,38)	JL	Silurian-Devonian*
Erquy Spilite Series N. Brittany	5	221	82	4.9	47	11	344E	35N	(21,22)	ES	Younger than folding, about 493 ± 10
Trégastel-Ploumanac'h granite complex	5	200	9	5.0	122	7	332E	34S	(4,7)	TP	300 ± 5
Leicestershire diorites	3	18	48	3.0	136	11	141E	63N	(9)	LD	~ 549
Armorican redbed component B3	7 (10spec)	216	41	6.8	36	10	324E	11S	(8,12)		Devonian - early Carboniferous*
Armorican redbed component C	7 (16spec)	228	63	6.9	71	7	323E	11N	(9,11)		Silurian - early Devonian*
Bangor red shales, (Lower Cambrian) North Wales	5 samples (30spec)	206	-40	4.7	15	21	132E	54N		BS	early Cambrian

* Age estimated by comparison of palaeomagnetic poles with the apw path for Europe; notation as for Table 1.

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Table 3. European palaeomagnetic poles for the period early Cambrian to early Permian.

Period	N	K	A_{95}	Mean south pole		Poles used ⊕
late Carboniferous (Stephanian) to early Permian	9	258	3	39S	354E	5.30 - 5.38
middle to late Carboniferous (pre Stephanian)	21	56	4	35S	335E	5.9 - 5.29
early Carboniferous	8	49	8	18S	333E	5.1 - 5.8
Silurian to Devonian	12	29	8	3N	321E	4.1 - 4.12
Ordovician	9	40	8	9S	5E	3.5, 3.11 - 3.18
early to middle Cambrian*	5	113	7	79N	2E	2.4 - 2.8
early Cambrian	3	205	9	60N	139E	2.1 - 2.3

* Owing to the large apparent polar movement in the Cambrian and the paucity of data for this period it has not been possible to estimate a mean late Cambrian pole. The few middle or late Cambrian poles are intermediate between the early and middle Cambrian groupings and the Ordovician pole and are positioned as follows: Jersey Volcanics (533±16 m.y.): 52N 323E; Mona complex (post tectonic): 42N 324E and Erquy Spilite Series (post tectonic); 35N 344E.

⊕ Numbers refer to poles in a listing available from the author on request.

N is the number of poles averaged; K is the precision of N poles; A_{95} is the circle of 95% confidence about the mean pole.

used. A listing of all poles used to construct the European (and North American) apw path is available from the author on request.

The new results from the Armorican Massif help define the Cambrian segment of the apw path. Means of polar groupings confirm the Ordovician to Carboniferous segments of the published apw path for Britain and lie on the same path as the Norwegian pole positions.

3.1.2 Precambrian to Cambrian

A Precambrian south apw path for Europe is tentatively suggested in Fig. 8(a) based on the Precambrian palaeomagnetic data from the Baltic Shield and Britain, including the Channel Islands. The pole moves from the southern hemisphere first north-east towards the equator then northwards across the equator to a position defined by poles from the 600 Myr Mona Complex (Piper 1976), the Longmynd dykes (Lomax & Briden 1977), the *in situ* or dip-corrected St. Peter Port Gabbro (E. Hailwood, private communication) of Late Precambrian to early Cambrian age, the Jersey group C dolerite dykes, and the Nexø Sandstone (Prasad & Sharma 1978). This is followed by a large apparent polar shift to the early Cambrian group.

The most novel feature of the apw path is that an early Cambrian quasi-static interval is followed by a large polar shift. The quasi-static interval is defined by the poles from the Fen complex of Norway (Poorter 1972), Leicestershire diorites, and a preliminary result from red shales in the Lower Cambrian of North Wales (Duff 1978b). This last result was based on five samples collected by M. P. Coward from Bangor (lat. 53.3° N; long. 356° E) which gave the following preliminary results: $D = 206°$; $I = -40°$; N = 5; $R = 4.73$; $\kappa = 15$; $\alpha_{95} = 21°$. The corresponding VGP is at 132° E 54° N. The pole then appears to have moved rapidly via high latitudes (defined by the Jersey granites, the Uriconian and Malvernian, and possibly the Longmynd sediments) through intermediate latitudes (defined by the Jersey Volcanics, the *in situ* Mona Complex and the Erquy Spilite Series) and thence towards low latitudes and on the Ordovician segment of the apw path (Fig. 8(b)). Rb–Sr whole rock results from the Jersey Andesite Formation (Duff 1978a) suggest that this shift is, at least partly, middle Cambrian in age. Owing to the lack of palaeomagnetic data for the period middle Cambrian to Ordovician, the detailed nature of the apw path between the Erquy

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Spilite Series pole (Duff 1979a) and the mean Ordovician pole is not clear (dashed line Fig. 8(b)).

3.1.3 Ordovician to Devonian

The Ordovician and Siluro–Devonian poles define two separate low-latitude groupings (Table 3) as in the published British apw path (e.g. Briden *et al.* 1973). The pole remained stationary with respect to Europe during the Ordovician. Most results from Ireland have not been used in the determination of the mean Ordovician pole (Table 3) owing to evidence of tectonic rotations within Ireland (Briden *et al.* 1973). Movement of the pole from the Ordovician to the Siluro–Devonian positions could correspond to a rotation of about 30° about a local axis (Morris 1976). Poles from the Jersey group B dolerites and lamprophyres, and the VGP corresponding to component C in the Armorican redbeds (Duff 1979a) are all positioned close to the mean Siluro–Devonian pole.

3.1.4 Late Devonian to late Carboniferous

Movement of the pole from its Siluro–Devonian to middle Carboniferous positions during the late Devonian or early Carboniferous may be recorded in lavas and sediments from Britain, Norway and the Armorican Massif. The simplest *prima facie* explanation of the discrepancy between poles from British and Norwegian Siluro–Devonian rocks is that the Norwegian (Old Red Sandstone) redbeds were remagnetized during the early Carboniferous. This accords with the position of the Norwegian poles close to those from British early Carboniferous rocks (see also McElhinny 1973). However, the discrepancy may alternatively be reconciled by Devonian relative movements between the Baltic Shield and Britain (Briden *et al.* 1973). Analysis of the multi-component NRM of the Armorican redbeds (Duff 1979a) suggests that they too were remagnetized during the late Devonian or early Carboniferous.

Two distinct polar groupings defined by (i) middle to late Carboniferous and (ii) late Carboniferous (Stephanian) to early Permian rocks, imply a polar shift during the early part of the late Carboniferous (Fig. 8(b)). The sources of both these groups of data are spread geographically throughout Europe. This shift, like that in the Ordovician to Siluro–Devonian positions, corresponds to a declination change and hence could have originated by rotation of the Palaeozoic European platform about a Euler pole close to, or on it. The declination change shows up well in the redbeds of Normandy and Brittany (Duff 1979a) and in the Bohemian Massif (Birkenmajer, Krs & Nairn 1968), owing to the near-horizontal Carboniferous field direction in Europe. Poles from the Jersey goup A dolerites and the Trégastel–Ploumanac'h complex (Duff 1979a) are positioned with the middle to late Carboniferous group (Fig. 8(b)).

3.2 PALAEOMAGNETIC DATING

The apw path permits the ages of certain rock units hitherto of uncertain or unknown age to be estimated. Jersey group C dolerite and Longmynd dyke pole positions are close to those for the *in situ* or dip-corrected St. Peter Port Gabbro of Guernsey, which is 660 to 570 Myr old (Adams 1976), and the 600 Myr Mona Complex (Piper 1976). This implies a Late Precambrian to early Cambrian age for the Jersey group C dykes and Longmynd dykes.

Palaeomagnetic poles from the Jersey granites fall on the early to middle Cambrian (pre 533 Myr) segment of the apw path, which implies that the Rb–Sr ages of the north-west (490 ± 15 Myr) and south-east (520 ± 4 Myr) granites may underestimate the age of intrusion. Intrusion of the south-east granite by the group C dolerites with their probable Late Precambrian to early Cambrian magnetization, implies that the south-east granite was

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intruded no later than early Cambrian, and that the characteristic NRM of the granites was acquired after group C dyke emplacement.

The position of poles determined from post-folding remagnetizations in the Uriconian and Malvernian of Britain (Lomax 1975) on the apw path (Fig. 8) implies that these rocks were remagnetized during the early to middle Cambrian. The position of the 'remagnetization pole' from the Longmynd sediments (LS) may indicate a slightly younger age of remagnetization in these sediments.

Comparison of palaeomagnetic poles for group B dolerites and lamprophyre dykes with the apw path suggests that these dykes are probably at least Silurian to middle Devonian in age. This is consistent with a hornblende K–Ar age of 427 ± 13 Myr obtained from a lamprophyre which intrudes the Rozel Conglomerate (Adams 1967). The Jersey dolerite group A pole position is very close to the 300 ± 5 Myr pole from the Trégastel–Ploumanac'h granite complex of northern Brittany (Duff 1979a), and to the middle to late Carboniferous pole positions from elsewhere in Europe (Fig. 8(b)). Accordingly, if NRM was acquired shortly after intrusion, group A dykes are of middle to late Carboniferous age, and were intruded during the Hercynian Orogeny.

3.3 LOWER PALAEOZOIC APPARENT POLAR WANDER

Reliable Lower Palaeozoic pole positions for North America have been compiled from the Geophysical Journal pole lists and from Van der Voo, French & Willians (1976), Elston & Bressler (1977) and J. C. Briden (personal communication). The means of polar groupings are shown in Table 4. There have been few recent Carboniferous palaeomagnetic studies, and the mean pole ($N = 13$) of McElhinny (1973) has been used for this period. Means of natural polar groupings were determined for the Lower Palaeozoic of the Russian and Siberian Platforms (Table 5) using the results listed by McElhinny & Cowley (1977). The reliable poles for the Lower Palaeozoic of Australia and Africa were taken from Hailwood (1974).

The Lower Palaeozoic apw paths for the Russian and Siberian Platforms, North America, Africa and Australia are compared with the European apw path in the equal area projection of Fig. 9. The apw paths are rotated to their position relative to pre-Mesozoic Pangaea using the Euler poles and rotation angles of Smith (1971) for Africa to Europe and of Smith

Table 4. North American palaeomagnetic poles for the period Late Precambrian (Hadrynian) to Carboniferous.

Period	N	K	A_{95}	Mean South Pole unrotated		rotated**		Poles used (not used)⊕
Carboniferous*	13	89	4	37S	306E	38S	334E	
Devonian (3)	4	144	8	56S	286E	57S	324E	4.9 - 4.12
Devonian (2)	3	1904	3	47S	298E	48S	337E	4.6 - 4.8
Silurian - Devonian (1)	4	35	16	26S	302E	27S	330E	4.1 - 4.4 (4.5)
Ordovician	3	372	6	36S	301E	37S	340E	3.1 - 3.3
Hadrynian - Cambrian	8	112	5	4S	341E	4S	19E	2.1 - 2.8

* The mean Carboniferous pole of McElhinny (1973)

** Poles are corrected to their pre-Mesozoic positions using the rotation pole of Bullard et al. (1965).

⊕ Numbers refer to poles in a listing available from the author on request.
Symbols as in Table 3.

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Table 5. Russian and Siberian Platform palaeomagnetic poles for the period early Cambrian to Carboniferous.

REGION	PERIOD	N	K	A_{95}	Mean South Pole		Poles Used*
RUSSIAN PLATFORM	Devono-Carboniferous	23	59	4	36S	342E	8.01-8.03, 8.05-8.09, 8.11-8.15, 8.35, 8.51-8.52, 9.01-9.05, 9.16-9.17
	Devonian	4	250	6	6S	352E	9.20, 9.22, 9.27, 9.37
SIBERIAN PLATFORM	Devono-Carboniferous	13	30	8	31S	320E	8.25-8.27, 8.36, 8.38, 8.45, 8.47, 8.50 9.19, 9.26, 9.32, 9.33, 9.35
	middle - late Ordovician	11	192	3	22N	311E	11.2-11.8, 11.12-11.15
	late Cambrian - early Ordovician	22	35	5	38N	310E	11.16-11.20, 12.3-12.9, 12.17-12.23, 12.25-12.27
	early - middle Cambrian	10	39	8	44N	339E	12.10-12.14, 12.24, 12.28-12.30, 12.32

* Pole reference numbers are those used in 'Palaeomagnetic results from the USSR' (McElhinny and Cowley, 1977)

Symbols as in Table 3.

& Hallam (1970) for Australia and Africa. The Euler pole of Bullard, Everett & Smith (1965) was used to rotate the North American poles to their pre-Mesozoic positions.

The apw paths differ on this Pangaea base map (Fig. 9), which implies that Europe, the Russian and Siberian Platforms, North America and Gondwanaland drifted independently during the Lower Palaeozoic. The Late Precambrian (Hadrynian) and Cambrian poles for North America are positioned close to the European Ordovician group as noted by Briden

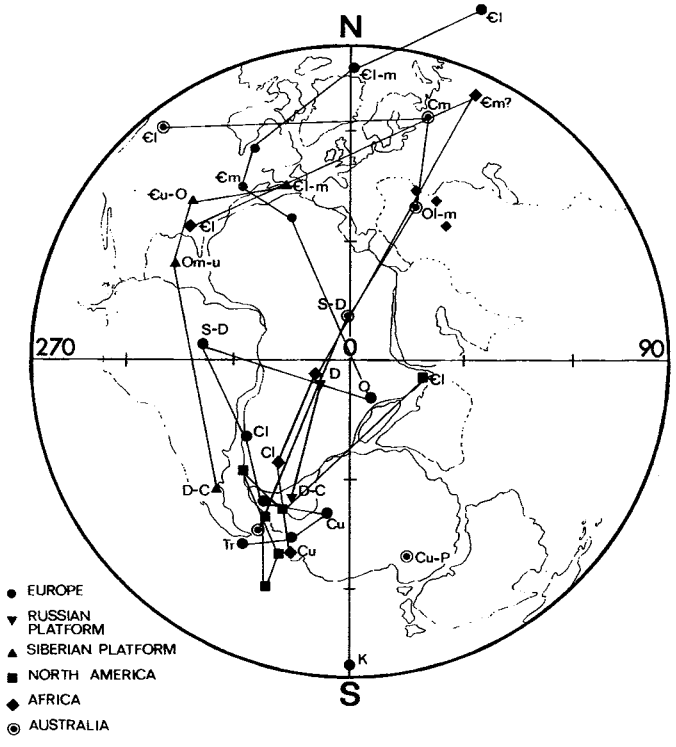


Figure 9. Apw paths for Europe, the Russian and Siberian Platforms, North America, Africa and Australia rotated to their positions relative to pre-Mesozoic Pangaea. Equatorial equal area projection.

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et al. (1973). They are separated by a large apparent polar shift from the North American Ordovician to Carboniferous poles (Table 4, Fig. 9), which are positioned near the European middle Carboniferous group and the Devono–Carboniferous mean poles for the Russian and Siberian Platforms. The convergence of all paths by the late Carboniferous reflects the formation of Pangaea.

3.4 PALAEOLATITUDES OF EUROPE DURING THE LOWER PALAEOZOIC

The mean poles which define the apw path for Europe (Table 3) have been used to determine the palaeolatitudes of Europe during the Lower Palaeozoic (Fig. 10). There is strong faunal province, palaeoclimatic and stratigraphic evidence that north-west Africa, at least as far south as the Anti-Atlas belt, was contiguous with Europe (Section 3.4.1). As its detailed palaeoposition south of Europe is not known, north-west Africa is shown in its present-day position relative to Europe (Fig. 10).

During the early Cambrian (Fig. 10(a)) Europe lay in low or temperate latitudes with north-west Africa close to the equator. This is consistent with early Cambrian warm water indicators in north-west Africa (carbonate deposits and abundant archaeocyathea; Jell 1974), and supports the contiguity of north-west Africa with Europe rather than Gondwanaland, for which near polar latitudes are predicted for northern Africa (Fig. 9; McElhinny, Giddings & Embleton 1974). By the early to middle Cambrian Europe had drifted much closer to the South Pole; the Baltic Shield lay in latitudes 65–80° S (Fig. 10(b)). The palaeolatitudes

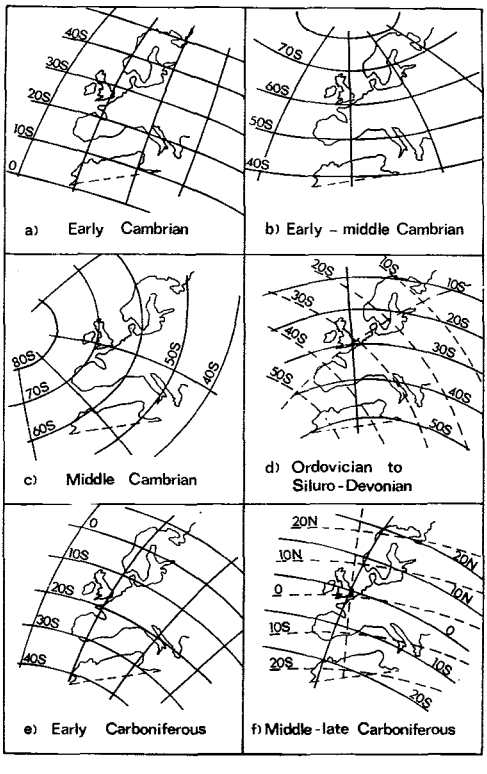


Figure 10. Palaeolatitudes of Europe for the early Cambrian to late Carboniferous. Rotations of Europe during the Ordovician to Devonian (d) and late Carboniferous (f) are indicated by the corresponding change from solid to dashed palaeolatitudes. Equal area projections.

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based on the Jersey Volcanics pole (JV) imply that during the middle Cambrian, south present-day Europe and north-west Africa rotated closer to the South Pole (Fig. 10(c)). This could accord with the evidence of glaciation in the Dalradian tillites, which are probably partly Cambrian in age (Bennison & Wright 1969).

During the middle to late Cambrian, Europe moved around and away from the South Pole toward low latitudes. By the Ordovician (Fig. 10(d)) the Baltic Shield and Britain were within about 30° of the equator and north-west Africa was positioned in intermediate latitudes. Europe rotated anticlockwise between the Ordovician (solid grid Fig. 10(d)) and Siluro–Devonian (dashed grid), and during the early Carboniferous drifted from its Ordovician to Devonian latitudes further toward the equator (Fig. 10(e)). By middle to late Carboniferous, Britain and adjacent Europe lay on the equator. During early Late Carboniferous times (Westphalian to Stephanian) the bulk of Europe rotated clockwise. This rotation is shown in Fig. 10(f) as the corresponding change in direction of the palaeomeridian from middle–late Carboniferous (solid grid) to late Carboniferous (dashed grid).

3.5 THE HERCYNIAN OROGENY

3.5.1 Integrity of Europe

The European Lower Palaeozoic apw path provides a common path for regions from the Baltic Shield to at least as far south as the Armorican Massif. This suggests that the basement of Europe extended from the Baltic Shield as far south as the Armorican Massif at least as far back as the early Cambrian, although there may have been minor relative movements between the Baltic Shield and Europe during Siluro–Devonian times (Briden *et al.* 1973). Rapid drift during the middle and possibly late Cambrian is implied for this integral European block by the disparate apw paths of Europe, Russia and North America and the Lower Palaeozoic palaeolatitudes. This was preceded by a period during the early Cambrian when the pole remained essentially stationary relative to Europe (quasi-static interval).

A single Lower Palaeozoic Europe as extensive as present day Europe is consistent with faunal province and stratigraphic evidence. Recent faunal province studies (Boucot, Johnson & Talent 1969; Hallam 1973; Jell 1974) strongly suggest that Europe from the Baltic Shield at least as far south as the Atlas belt of north-west Africa was an integral block, and this is consistent with the continuity of the Ordovician Grès Armorican orthoquartzite facies from Devon and Cornwall through the Armorican massif and Spain to Morocco (Renouf 1974; Ager 1975). Detailed stratigraphic and faunal studies in the Armorican Massif and southern England indicate that these regions were contiguous during the Lower Palaeozoic (Renouf 1974).

Although the Cambrian palaeomagnetic poles of north-west Africa (Helsley 1965; Ileana 1971; Daly & Pozzi 1977; poles shown as small diamonds in Fig. 9) may be interpreted as falling close to the poorly defined Cambro–Ordovician segment of the African apw path, it would be equally possible to reconcile them with the European path if their directions of NRM have been tectonically rotated from their original declinations during the Hercynian and Alpine deformations of north-west Africa (e.g. Sougy 1962). Early to middle Cambrian palaeolatitudes for Europe and north-west Africa are mutually compatible.

The palaeomagnetic and geological indications of an integral European–north-west Africa block conflict with most plate tectonic models of the Hercynian Orogeny (e.g. Burrett 1972; Dewey & Burke 1973; Johnson 1973; Laurent 1972; McKerrow & Ziegler 1972) which relate the acme of orogeny during the Westphalian to collision of north and south Europe, during final closure of a mid-European sea (Whittington & Hughes 1972). These models interpret Lower Palaeozoic deep water sediments and pillow lavas in southern England, the

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Ardennes and the Armorican Massif as continental margin suites, contrary to the intra-continental interpretations of detailed studies (e.g. Floyd 1972; Renouf 1974).

3.5.2 Intracratonic orogeny

The palaeomagnetic and geological evidence of an integral Lower Palaeozoic Europe–north-west Africa block and the location of the Hercynides in Europe, are most consistent with an intra-continental setting for the Hercynian Orogeny, although the late Carboniferous climactic phase may have been associated with the fusion of Europe, North America and Gondwanaland at remote continental margins. This is contrary to interpretations which regard the Hercynian Orogeny as a continental margin type; faunal province studies suggest that any plate margin which separated Europe and Gondwanaland should be sought south of the Anti-Atlas belt. The relative positions of Europe–north-west Africa and the remainder of Africa during the early Carboniferous (Fig. 9, Briden, Drewry & Smith 1974) imply that if there was any closure between the two regions during the middle to late Carboniferous, it was oblique (transform) rather than direct convergence.

An intracratonic interpretation of Lower Palaeozoic orogeny in Europe coincides with the views of Zwart (1969) and Krebs & Wachendorf (1973), who emphasize the distinctive features of the Hercynian Orogeny, especially its widespread, diffuse and protracted nature. Zwart (1969) suggests that 'hercynotype' orogeny occurs when continental crust drifts over mantle with anomalously high heat flow.

3.5.3 Rotation of Europe during the Hercynian Orogeny

Because the poles defining the distinct middle to late and late Carboniferous groupings are derived from regions throughout Europe, it is considered that Europe rotated clockwise (about 15°) as an integral block. It is unnecessary, therefore, to invoke differential local rotations within Europe during the late Carboniferous to early Permian to explain the declination scatter of Carboniferous NRM (e.g. Birkenmajer *et al.* 1968), since such scatter in the Armorican Massif (Duff 1979a) and Bohemian Massif NRM may be explained by the same, coherent rotation of all Europe.

This rotation and the climactic Westphalian fold deformation may have been both related to oblique movement of the African part of Gondwanaland toward Europe; the resulting right–lateral shear may have had a 'roller ball' effect on Europe which responded by clockwise rotation. This transcurrent relative movement and clockwise rotation is consistent with structural evidence of relative right–lateral displacement between Europe and Africa during the late Carboniferous on a megashear zone some 10 000 km long and 1000 km wide (Arthaud & Matte 1977). Palaeomagnetic observations also imply right–lateral shear displacement of some 3000 km between Europe and Africa during middle Triassic to Jurassic (Irving 1977).

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References

Adams, C. J. D., 1967. K/Ar ages from the basement complex of the Channel Islands (United Kingdom) and the adjacent French mainland, *Earth planet. Sci. Lett.*, **2**, 52–56.

Adams, C. J. D., 1976. Geochronology of the Channel Islands and adjacent French mainland, *J. geol. Soc. Lond.*, **132**, 233–250.

Ade-Hall, J. M., 1964. Electron probe microanalyser analyses of basaltic titanomagnetites and their significance to rock magnetism, *Geophys. J. R. astr. Soc.*, **8**, 301–312.

Ade-Hall, J. M., Wilson, R. L. & Smith, P. J., 1965. The petrology, Curie Points and natural magnetizations of basic lavas, *Geophys. J. R. astr. Soc.*, **9**, 323–336.

Ager, D. V., 1975. The geological evolution of Europe, *Proc. Geol. Ass.*, **86**, 127–154

Arthaud, F. & Matte, P., 1977. Late Precambrian strike–slip faulting in southern Europe and northern Africa: result of a right–lateral shear zone between the Appalachians and the Urals. *Bull. geol. Soc. Am.*, **88**, 1305–1320.

Bennison, G. M. & Wright, A. E., 1969. *The geological history of the British Isles*, Edward Arnold, London.

Birkenmajer, K., Krs, M. & Nairn, A. E. M., 1968. A palaeomagnetic study of Upper Carboniferous rocks from the Inner Sudetic Basin and the Bohemian Massif, *Bull. geol. Soc. Am.*, **79**, 589–608.

Boucot, A. J., Johnson, J. G. & Talent, J. A., 1969. Early Devonian brachiopod zoogeography, *Spec. Pap. geol. Soc. Am.*, 119.

Briden, J. C., Irons, J. & Johnson, P. A., 1970. Palaeomagnetic studies of the Caerfai Series and the Skomer Volcanic Group (Lower Palaeozoic, Wales), *Geophys. J. R. astr. Soc.*, **22**, 1–16.

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides – VI Regional and global implications, *Geophys. J. R. astr. Soc.*, **34**, 107–134.

Briden, J. C., Drewry, G. E. & Smith, A. G., 1974. Phanerozoic equal-area world maps, *J. Geol.*, **82**, 555–574.

Bullard, E., Everett, J. E. & Smith, A. G., 1965. The fit of the continents around the Atlantic, *Phil. Trans. R. Soc. Lond.*, **A258**, 41–51.

Burrett, C. F., 1972. Plate tectonics and the Hercynian Orogeny, *Nature*, **239**, 155–157.

Cox, A., 1969. Confidence limits for the precision parameter K, *Geophys. J. R. astr. Soc.*, **18**, 545–549.

Creer, K. M., Hedley, I. G. & O'Reilly, W., 1975. Magnetic oxides in geomagnetism, in *Magnetic oxides*, ed. Craik, D. J., J. Wiley.

Daly, L. & Pozzi, J. P., 1977. Détermination d'un nouveau pôle palaeomagnétique africain sur des formations cambriennes du Maroc, *Earth planet. Sci. Lett.*, **34**, 264–272.

Dewey, J. F. & Burke, K. C. A., 1973. Tibetan, Variscan, and Precambrian basement reactivation: products of continental collision, *J. Geol.*, **81**, 683–692.

Duff, B. A., 1978a. Rb–Sr whole-rock age determinations of the Jersey Andesite Formation, Jersey, C.I., *J. geol. Soc. Lond.*, **135**, 153–156.

Duff, B. A., 1978b. Palaeomagnetic and rock magnetic studies of Lower Palaeozoic rocks on Jersey and the adjacent regions of the Armorican Massif. *PhD thesis*, 338 pages, University of Leeds.

Duff, B. A., 1979a. The palaeomagnetism of Cambro–Ordovician redbeds, the Erquy Spilite Series and the Trégastel–Ploumanac'h granite complex, Armorican Massif (France and the Channel Islands), *Geophys. J. R. astr. Soc.*, **59**, 345–365.

Duff, B. A., 1979b. Palaeomagnetism of Late Precambrian or Cambrian diorites from Leicestershire, *Geol. Mag.*, in press.

Elston, D. P. & Bressler, S. L., 1977. Palaeomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, *Earth planet. Sci. Lett.*, **36**, 423–433.

Eugster, H. P. & Wones, D. R., 1962. Stability relations of the ferruginous biotite, annite, *J. Petrol.*, **3**, 82–125.

Floyd, P. A., 1972. Geochemistry, origin and tectonic environment of the basic and acidic rocks of Cornubia, England, *Proc. Geol. Ass.*, **83**, 385–404.

Hailwood, E. A., 1974. Palaeomagnetism of the Msissi Norite (Morocco) and the Palaeozoic reconstruction of Gondwanaland, *Earth planet. Sci. Lett.*, **23**, 376–386.

Hallam, A., 1973. *Atlas of Palaeobiogeography*, Elsevier.

Helsley, C. E., 1965. Palaeomagnetic results from the Middle Cambrian of north-west Africa (abstract), *Trans. Am. Geophys. Un.*, **46**, 67.

Ileana, B., 1971. A palaeomagnetic study of a sedimentary formation from the Algerian Sahara of Cambro–Ordovician age (abstract) first European Earth Planet. Phys. Colloquium, Reading.

Irving, E., 1964. *Palaeomagnetism and its Application to Geological and Geophysical Problems.* Wiley, New York.

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Irving, E., 1977. Drift of the major continental blocks since the Devonian, *Nature,* **270,** 304–309.

Jell, P. A., 1974. Faunal provinces and possible planetary reconstruction of the Middle Cambrian, *J. Geol.,* **82,** 319–350.

Johnson, G. A. L., 1973. Closing of the Carboniferous sea in western Europe, in *Implications of Continental Drift to the Earth Sciences*, eds Tarling, D. H. & Runcorn, S. K., Academic Press, London, 843–850.

Krebs, W. & Wachendorf, H., 1973. Proterozoic–Palaeozoic geosynclinal and orogenic evolution of central Europe, *Bull. geol. Soc. Am.,* **84,** 2611–2629.

Laurent, R., 1972. The Hercynides of South Europe, a model, *Proc. 24th Int. Geol. Cong.,* **3,** 363–370.

Lomax, K., 1975. Palaeomagnetic studies of Proterozoic rocks in Britain and West Africa, *PhD thesis,* University of Leeds.

Lomax, K. & Briden, J. C., 1977. Palaeomagnetic studies of the Longmyndian and other British Late Precambrian/early Palaeozoic rocks, and their regional tectonic implications, *J. geol. Soc. Lond.,* **133,** 5–21.

McElhinny, M. W., 1973. *Palaeomagnetism and Plate Tectonics,* Cambridge University Press.

McElhinny, M. W. & Briden, J. C., 1971. Continental drift during the Palaeozoic, *Earth planet. Sci. Lett.,* **10,** 407–416.

McElhinny, M. W., Giddings, J. W. & Embleton, B. J. J., 1974. Palaeomagnetic results and late Precambrian glaciations, *Nature,* **248,** 557–561.

McElhinny, M. W. & Cowley, J. A., 1977. Palaeomagnetic results from the USSR, *RSES publication no. 1268,* Australian National University, Research School of Earth Sciences.

McKerrow, W. S. & Ziegler, A. M., 1972. Palaeozoic oceans, *Nature Phys. Sci.,* **240,** 92–94.

Morris, W. A., 1976. Transcurrent motion determined palaeomagnetically in the northern Appalachians and Caledonides and the Acadian Orogeny, *Can. J. Earth Sci.,* **13,** 1236–1243.

Mourant, A. E., 1933. The geology of eastern Jersey, *Q. Jl geol. Soc. Lond.,* **89,** 273–307.

Néel, L., 1948. Propriétés magnétiques des ferrites; ferrimagnétisme et antiferromagnétisme, *Ann. Physique,* **3,** 137.

Piper, J. D. A., 1976. Magnetic properties of Precambrian pillow lavas of the Mona Complex and a related dyke swarm, Anglesey, Wales, *Geol. J.,* **11,** 189–201.

Poorter, R. P. E., 1972. Preliminary palaeomagnetic results from the Fen Carbonatite complex, South Norway, *Earth planet. Sci. Lett.,* **17,** 194–198.

Prasad, S. N. & Sharma, P. V., 1978. Palaeomagnetism of the Nexø Sandstone from Bornholm Island, Denmark, *Geophys. J. R. astr. Soc.,* **54,** 669–680.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.,* **28,** 133–143.

Renouf, J. T., 1974. The Proterozoic and Palaeozoic development of the Armorican and Cornubian provinces, *Proc. Ussher Soc.,* **3,** 6–43.

Smith, A. G., 1971. Alpine deformation and the oceanic areas of the Tethys, Mediterranean, and Atlantic, *Bull. geol. Soc. Am.,* **82,** 2039–2070.

Smith, A. G. & Hallam, A., 1970. The fit of the southern continents, *Nature,* **225,** 139–144.

Sougy, J., 1962. West African fold belt, *Bull. geol. Soc. Am.,* **73,** 871–876.

Storetvedt, K. M. & Halvorsen, E., 1968. On the palaeomagnetic reliability of the Scottish Devonian lavas, *Tectonophysics,* **5,** 447–457.

Van der Voo, R., French, R. B. & Williams, D. W., 1976. Palaeomagnetism of the Wilberns Formation (Texas) and the late Cambrian palaeomagnetic field for North America, *J. geophys. Res.,* **81,** 5633–5638.

Watson, G. S. & Irving, E., 1957. Statistical methods in rock magnetism, *Mon. Not. R. astr. Soc. Geophys. Suppl.,* **7,** 289–300.

Whittington, H. B. & Hughes, C. P., 1972. Ordovician geography and faunal provinces deduced from trilobite distribution, *Phil. Trans. R. Soc.,* ser. B, **263,** 235–278.

Wilson, R. L. & Haggerty, S. E., 1966. Reversals of the Earth's magnetic field, *Endeavour,* **25,** 104–109.

Zwart, H. J., 1969. Metamorphic facies series in the European orogenic belts and their bearing on the causes of orogeny, *Geol. Ass. Canada Spec. Pap.* **5,** 7–16.

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